Long-Range Remote Lite: User Guide

Oct 4, 2026 · @Luke Jamieson

What it is

The Lite is a universal IR remote with a long reach. A head of 66 narrow-beam infrared LEDs sends the codes for TVs, amps, soundbars, projectors, set-top boxes, air-conditioners and more, much further than a normal remote. You pick your brand and model on the screen, and the six keys under it send the codes.

The LEDs run from four AA NiMH cells that you take out and charge in any NiMH charger. The screen and its processor run from their own 18650 cell, charged through USB-C. All the codes (about 233,000) live on a microSD card, so you update them by swapping one file.

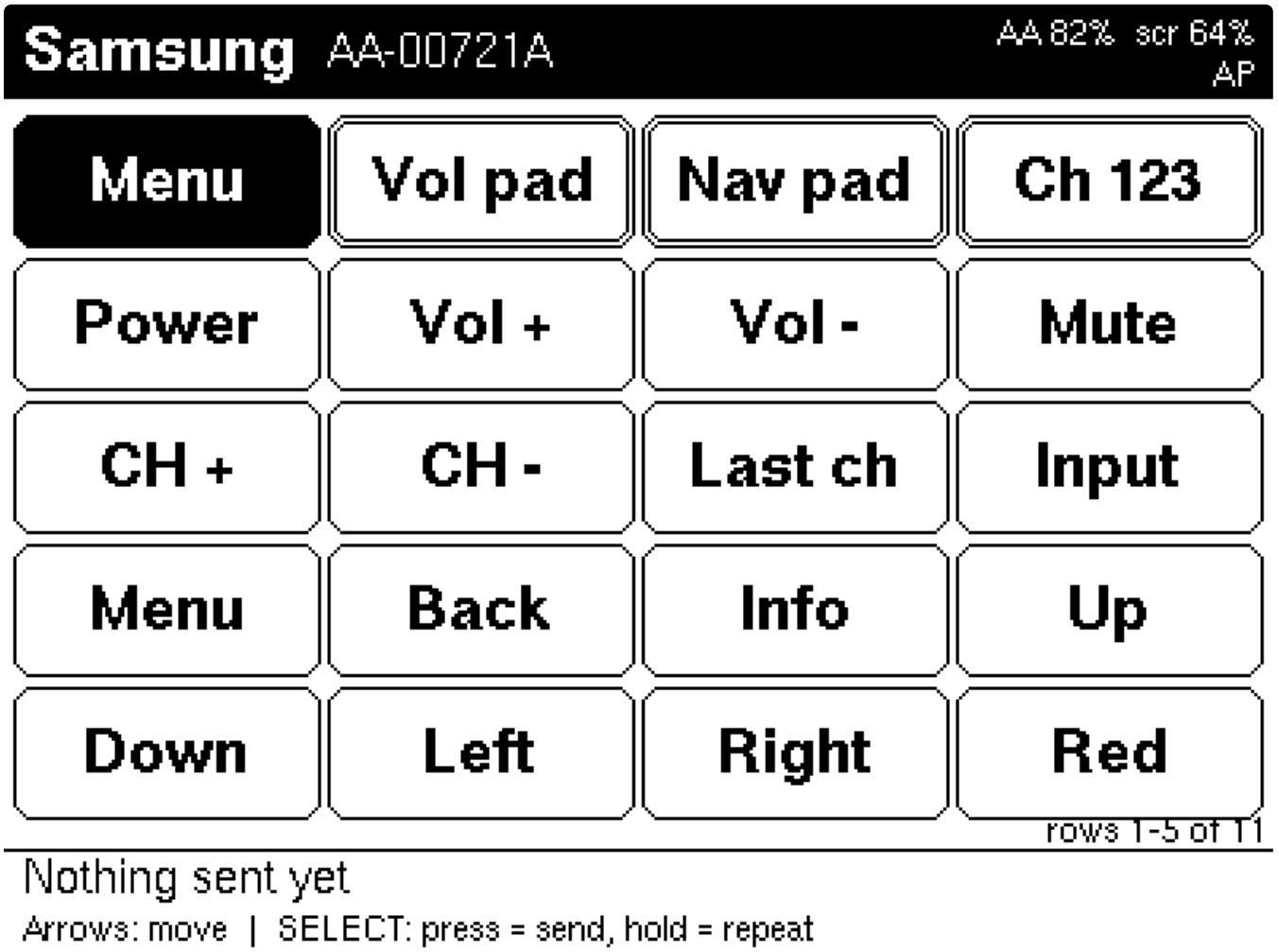
What you need to use it: the built remote, a FAT32 microSD card holding `irdb.bin`, four AA NiMH cells, a NiMH charger, and a USB-C cable for the screen's own battery.

First-time setup

1. **Load the codes.** Copy `irdb.bin` to the root of a FAT32 microSD card. Push the card into the TF slot in the right side of the case. Without the card the remote still starts, and the screen tells you what is missing.
2. **Charge the screen's battery.** Plug a USB-C cable into the port in the right side of the case. This charges the display board's own 18650 cell. The header shows `scr nn%` for its level.
3. **Put in the AA cells.** Undo the one screw in the battery door on the underside, lift the door out by the finger notch, and put four charged NiMH cells in the holder. Refit the door with its tongue in first, then swing the back down and tighten the screw. Use NiMH only: alkaline cells run the LEDs too hard.
4. **Turn the screen on** with the display board's own PWR button. It is reached through the slot in the lid just in front of the screen: drop a thin tool in beside the screen's front edge and push it rearwards onto the button. PWR is the middle one of the three buttons there; KEY (which also works as UP) is on its left and BOOT (which also works as SELECT) on its right. A short press turns the screen on, a long press turns it off.
5. **Check the header.** There is no power switch: the remote powers the LEDs itself, only while it sends. With four cells in, the header reads `AA nn%`; with none in, `AA out`.
6. **Pick your device.** Open the menu, choose Browse brands, then your brand and model. See Sending codes below.

The controls

The four triangle keys form a D-pad and each points the way it moves. The round key is SELECT. The square key at the D-pad's lower right is BACK: it goes one screen back from anywhere. Hold a key for fast repeat in lists.

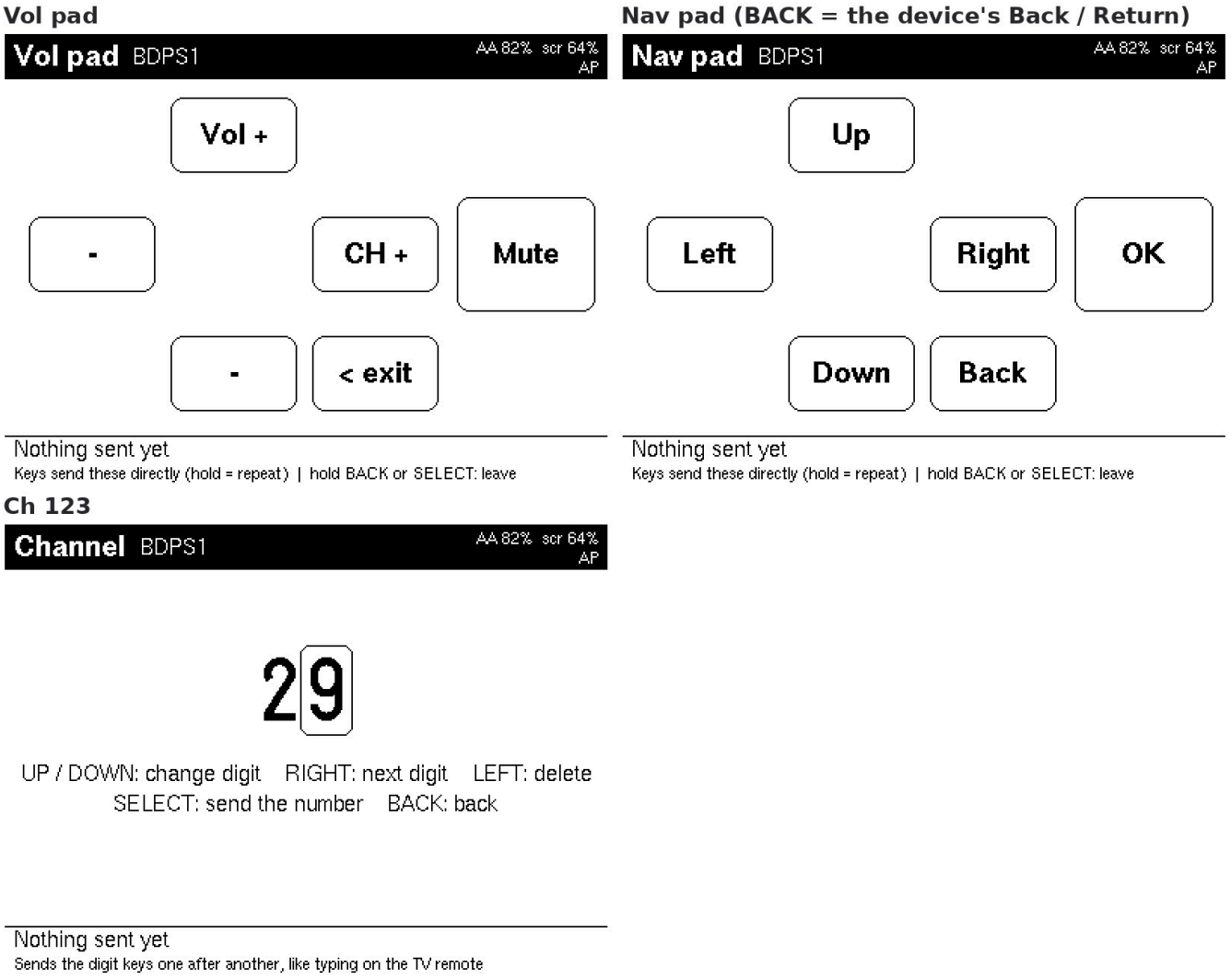


The remote screen. The header shows the brand, model, both batteries and AP (Wi-Fi on).

Screen	Arrow keys	SELECT
Lists (menu, brands, models)	UP/DOWN move (hold = fast). RIGHT chooses	Chooses. BACK goes back (so do LEFT and holding SELECT)
Remote screen (grid of keys)	Move around the grid	**Sends** the highlighted key. Hold to repeat (volume, channel)
Vol pad	UP/DOWN volume, LEFT/RIGHT channel − / +	Mute. BACK leaves
Nav pad	The device's arrows	OK. BACK sends the device's own Back / Return; hold BACK or SELECT to leave

Screen	Arrow keys	SELECT
Ch 123 (channel number)	UP/DOWN pick a digit, RIGHT adds it, LEFT deletes	Sends the number. BACK leaves
Code search	RIGHT next code, LEFT previous (each is sent)	Sends again. **Hold = "that worked, use it"**. BACK stops

The Vol pad and Nav pad let you aim and press without looking at the screen. They only appear when the device has those keys.

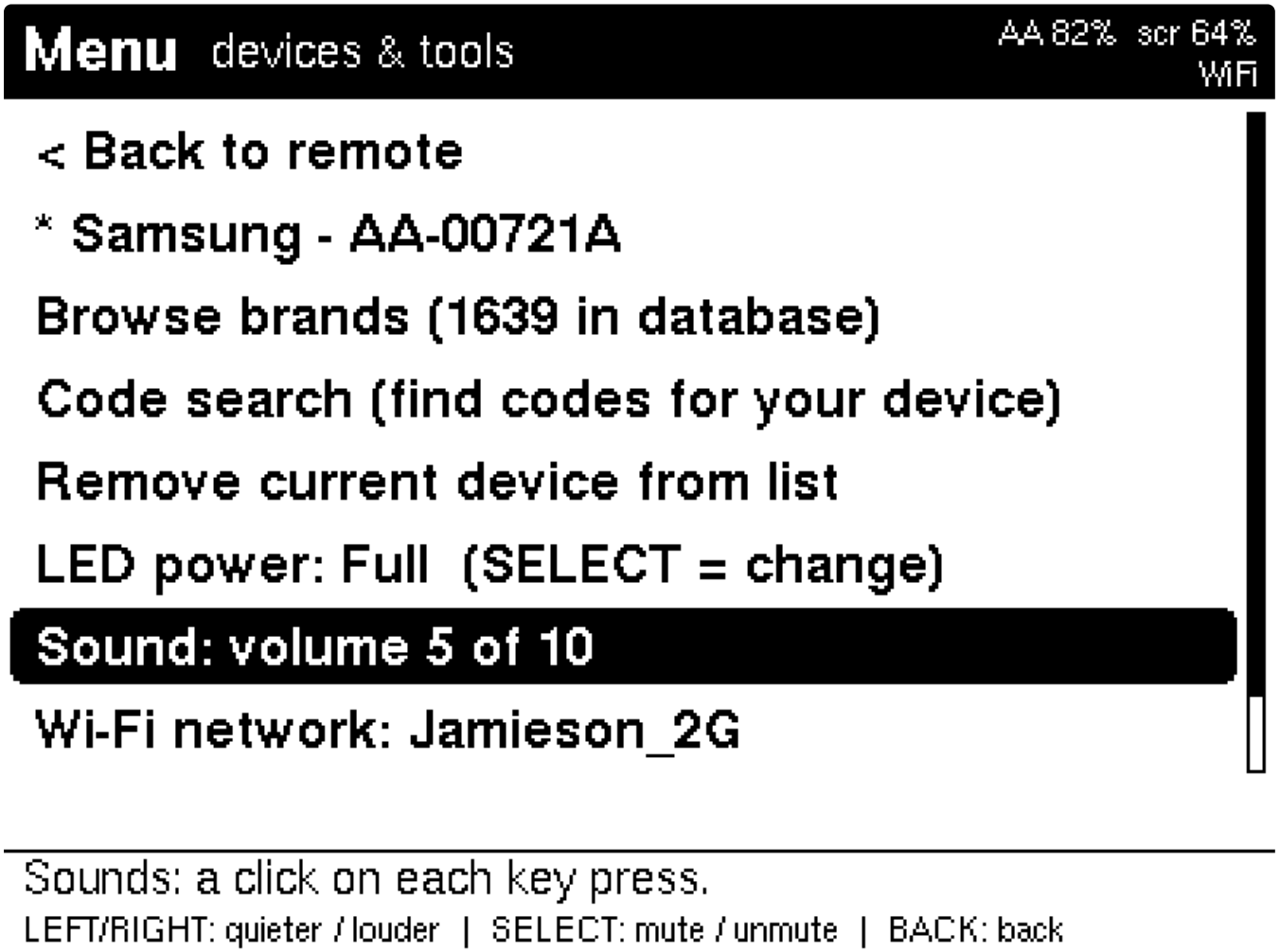


Vol pad (top left), Nav pad (top right) and Ch 123 (below). In the Nav pad, BACK sends the device's own Back key, handy in disc and streaming menus.

LED power: there is no power switch. Menu → LED power (SELECT cycles) or the web page's Settings tab sets Full, Half or Low. Use Full for long range; Half and Low save the cells at close range. The setting is remembered. The screen's own PWR button (long press) turns the screen off.

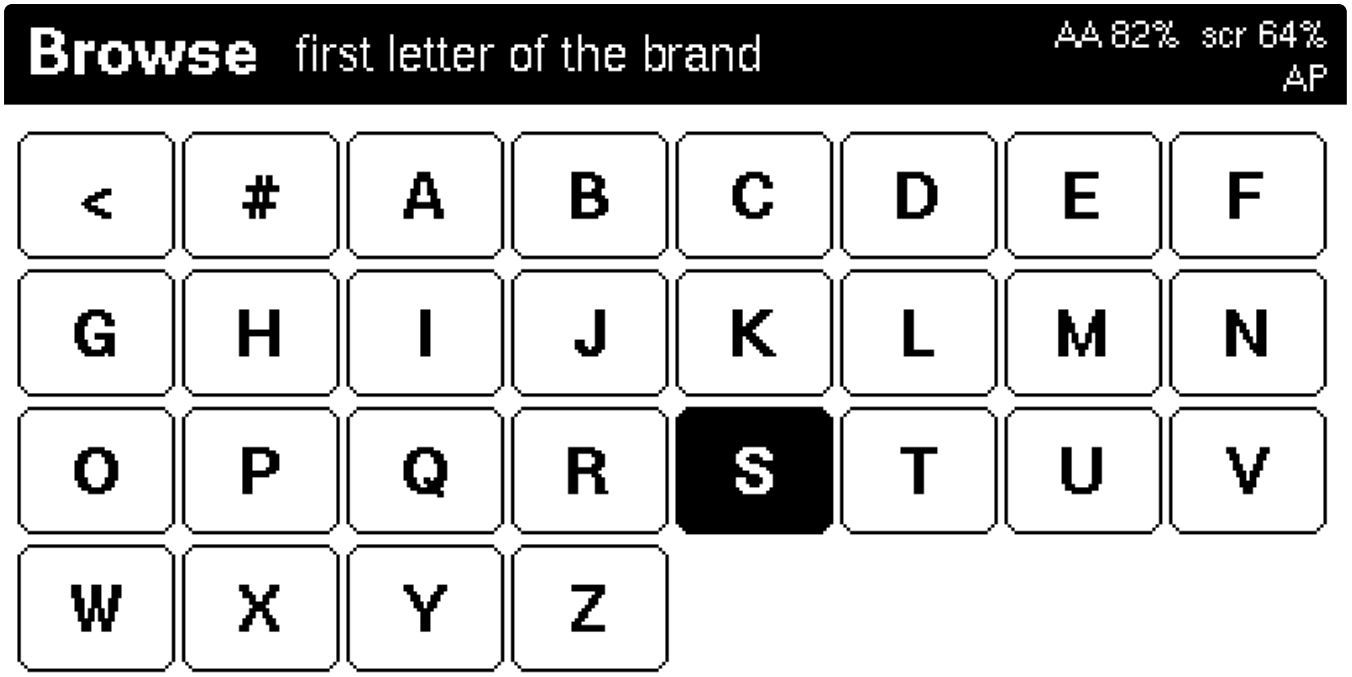
Sound: every key press gives a short click from the screen's speaker. Holding a key to leave a screen or pick a device gives a lower two-note sound, and a code that can't be sent because the AA cells are out or flat gives a low double buzz. To change the volume, open Menu → Sound: with that row selected, LEFT and RIGHT make it quieter or louder (10 steps, with a click at each), and SELECT mutes or unmutes it. Both are remembered.

Sending codes



Menu: your saved devices, Browse brands, Code search, LED power, Sound, Wi-Fi network and Status. With the Sound row selected, the footer shows its keys.

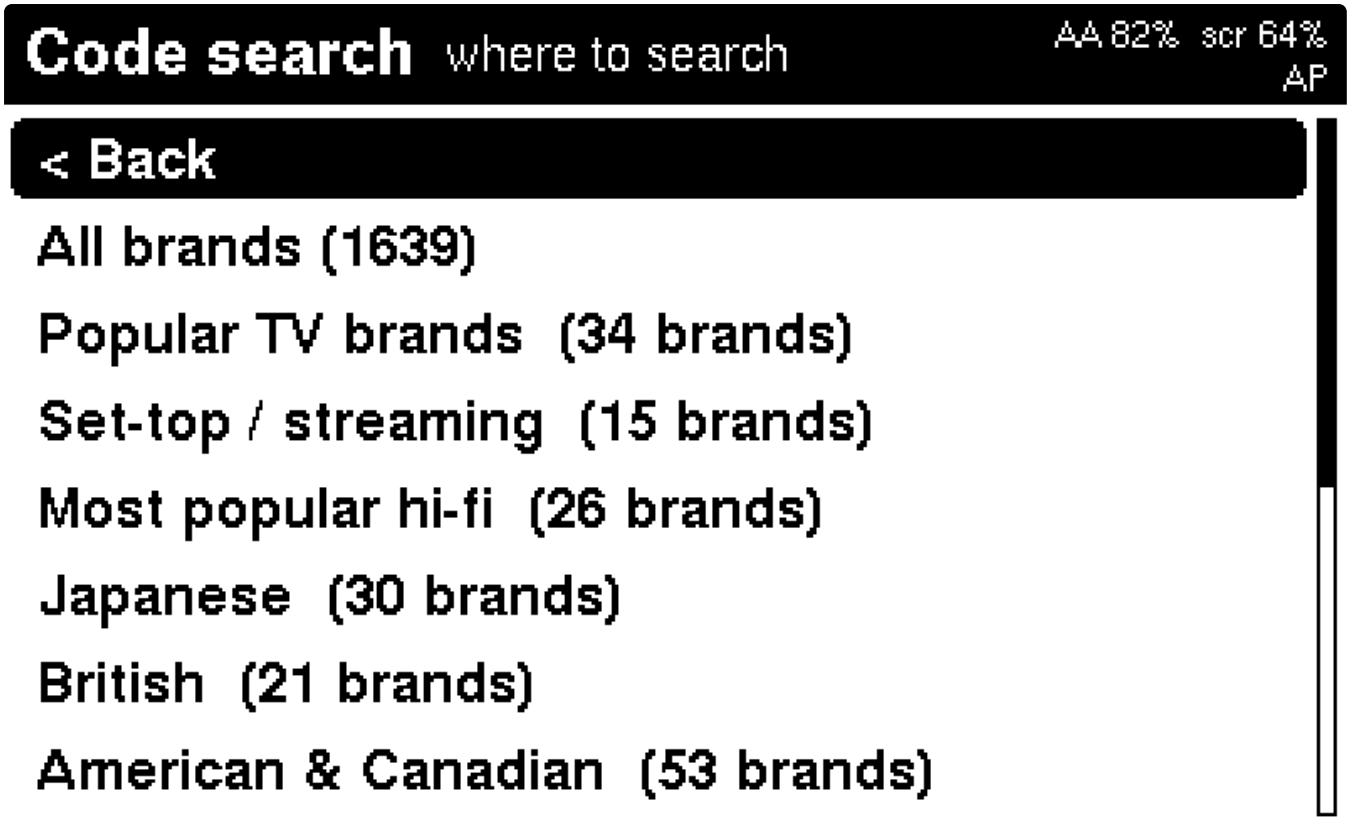
If you know the brand: Menu → Browse brands. Choose the first letter, then the brand, then the model. Model names come from real remotes. Entries shown as "TV 7,7" are code sets: device 7, sub-device 7. Pick the one that matches your device type.



Arrows: move | SELECT: choose | BACK: back

Browse brands starts with a letter grid. `<` *(or BACK) goes back,* `#` *is for brands starting with a digit.*

If you don't: Menu → Code search.



Code search starts by asking where to search.

1. Choose where to search: one brand, a group such as "Popular TV brands", or everything.
2. Choose the device type, then a test key such as Power, Vol + or Mute.
3. Aim at the device and step through the codes with RIGHT. Each one is sent as you land on it.
4. When the device reacts, **hold SELECT**. The remote saves it as your device.

The most widely shared codes come first, so a TV usually reacts within the first few tries. Remotes that share a code are only tried once.

Using your device: open it from Menu, where your saved devices are listed (`*` marks the active one). Move to a key and press SELECT to send it. Hold SELECT on volume or channel to repeat. Channel numbers go out 350 ms apart, because TVs ignore digits sent faster than a person types.

Your own remotes: copy Flipper Zero `.ir` files onto the card under `/ir`, for example `/ir/Living room/TV.ir`. They appear in the brand list as "Living room". Saved devices are stored by name, so they survive power-off and code updates.

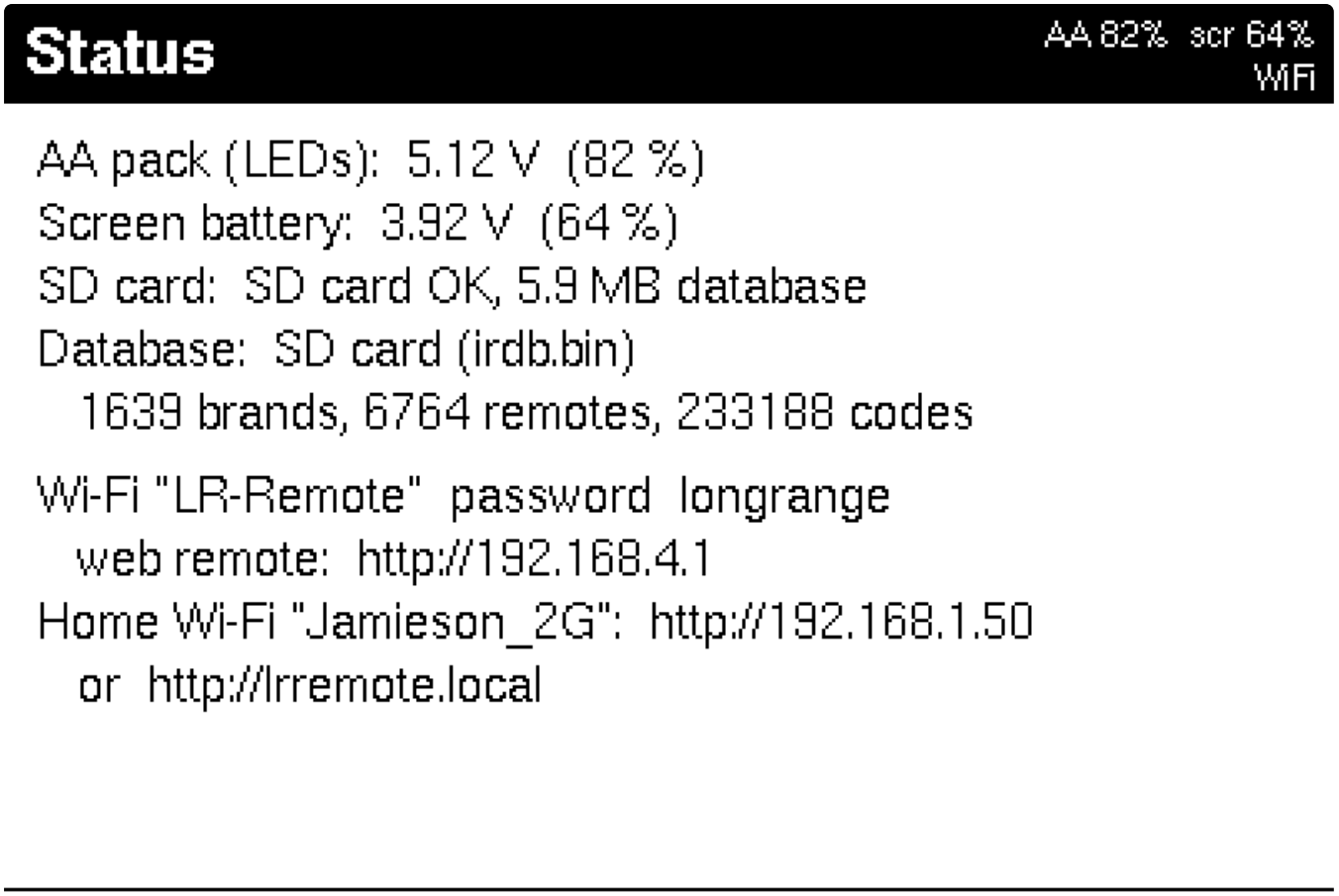
Range tip: point the head at the device, and keep the front clear. The bezel is level with the LED tips, so you can also rest the remote face-down on a table.

Batteries and charging

There are two separate batteries, and the header shows both, for example `AA 78% scr 64%`.

	AA pack (the LEDs)	Screen battery (the display and ESP32)
Cells	4 × AA NiMH, about 2000 mAh	One unprotected 18650 on the display board (the board has its own protection circuit; longer protected cells may not fit its holder)
Charge it	Take the cells out, charge in any NiMH charger	USB-C port in the right side, cells stay in
Header shows	`AA nn%`, or `AA out` with no cells in	`scr nn%`
Swap or charge when	The header says `LOW`, or below 4.6 V	The percentage gets low

- **Runtime:** a charged AA set gives about 2.3 hours of continuous sending, which is tens of thousands of key presses. Idle, the pack draws almost nothing (about 5 µA for the voltage reading).
- **Use NiMH, not alkaline.** Fresh alkaline cells read 6.0–6.4 V, which drives the LEDs past their rating. Low-self-discharge NiMH cells keep their charge in a drawer.
- **Range fades slowly.** It is strongest on freshly charged cells, holds steady through most of the discharge, then drops when the header shows `LOW`.
- **Flat cells stop sending.** When the cells are flat (4.4 V for the set, at rest), the remote stops sending and the footer says "AA cells flat". Swap in charged cells. This also protects the NiMH cells from over-discharge.
- **Sleep:** after 90 seconds idle the remote light-sleeps. Any key wakes it and is sent straight away. The screen keeps showing its last picture.
- **Storage:** nothing to switch off: the LEDs are disconnected from the pack whenever the remote isn't sending. For long storage, take the AA cells out.



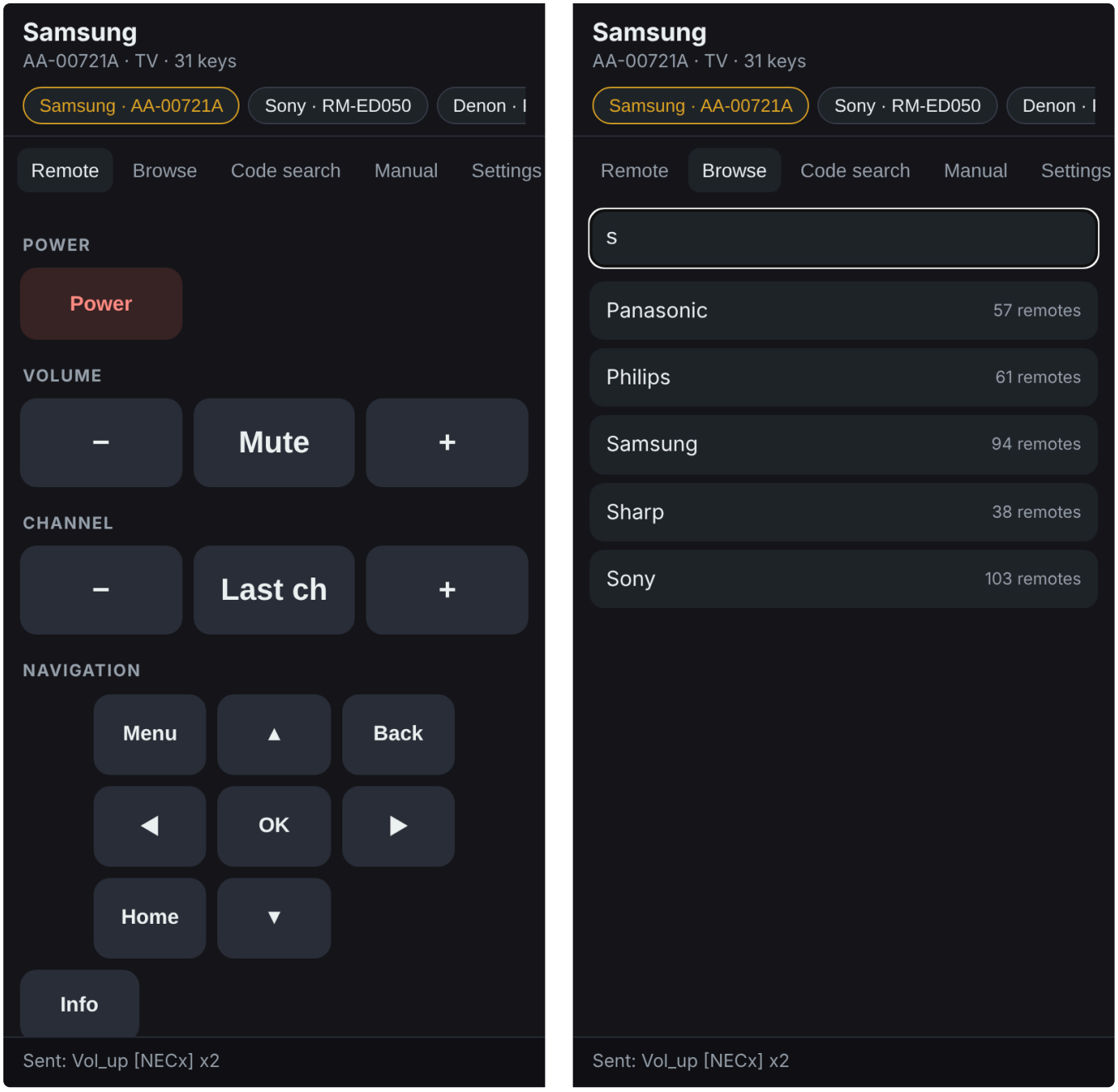
Menu → Status: both batteries, SD card, database and Wi-Fi details. Figures here are sample values.

The web page

You can run the remote from your phone, and update its codes there.

1. Wake the remote with any key.
2. On your phone, join the Wi-Fi network **LR-Remote** (password `longrange`).
3. Open **http://192.168.4.1** in the phone's browser.

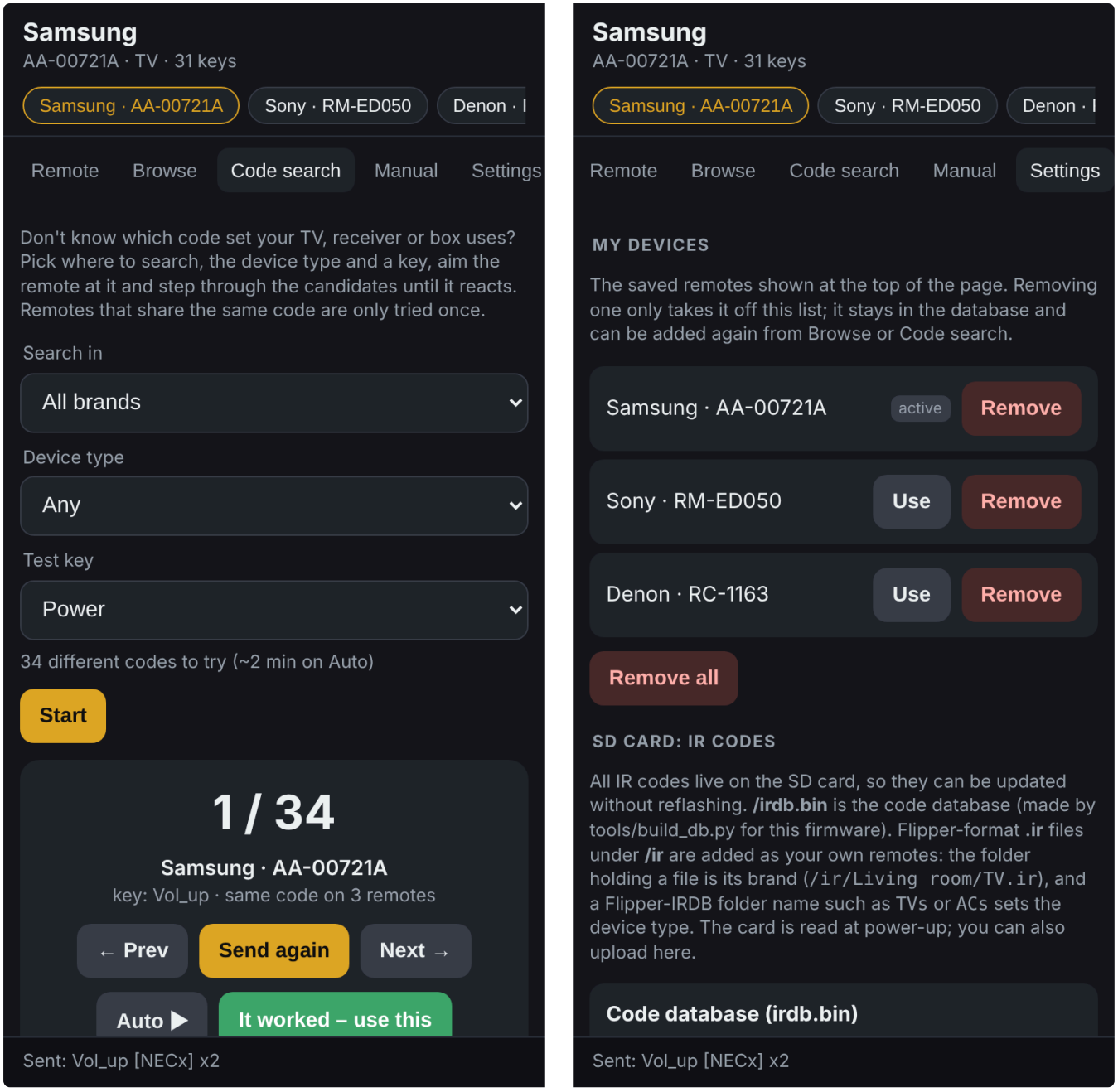
If the remote is on your home Wi-Fi as well (see **Home Wi-Fi** below), the page is also at **http://lrremote.local** from any phone on that network.



The web page on a phone: the Remote tab (left) and Browse (right). Sample data.

The Settings tab shows both battery voltages and lets you upload a new `irdb.bin` (the remote checks it, keeps the old one as `irdb.old`, and restarts) or your own `.ir` files.

Format the SD card (also on the Settings tab) erases the card and formats it FAT32, for a new card or one the remote can't read. Leave **Put the code database and home Wi-Fi back afterwards** ticked and the remote writes both back from its memory, so no upload is needed. Your own `.ir` files are erased with everything else, so keep copies. Formatting takes from a few seconds to about a minute.

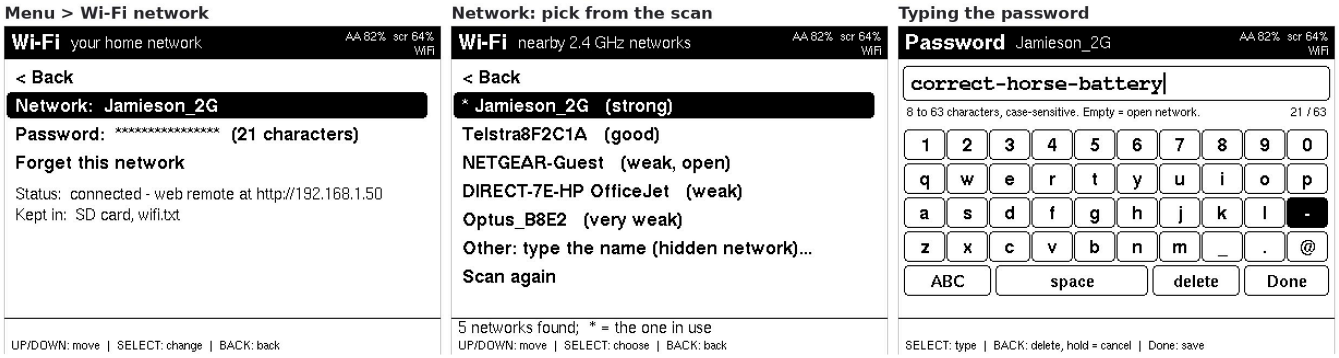


Code search (left) and Settings (right). Sample data.

Wi-Fi switches off when the remote sleeps, and the web page cannot wake it. Press a key first. While a phone stays joined to LR-Remote, the remote stays awake.

Home Wi-Fi

As well as running its own LR-Remote network, the remote can join your home network. Choose it on the remote: **Menu → Wi-Fi network**.



The Wi-Fi screen (left) shows the network, its connection status and where it's kept. Network scans for nearby networks (middle). Choosing a secured one opens the keyboard for its password (right).

- **Network** lists the 2.4 GHz networks nearby, strongest first; `*` marks the one in use. Pick yours, then type its password. An open network needs no password. For a hidden network, choose **Other** and type its name.
- **The keyboard:** the arrows move and SELECT types. BACK deletes a character; hold BACK to leave without saving. The bottom-left key switches between lower case, upper case and symbols. **Done** saves.
- **Password** changes the password of the current network. **Forget this network** (press SELECT twice) goes back to LR-Remote only.

Every change is saved to `wifi.txt` on the SD card straight away, and the remote reconnects. You can also edit that file on a computer. It is plain text:

```
ssid=MyNetwork
password=my wifi password
```

The remote creates `wifi.txt` the first time it starts with the card in. If there's no card, a change made in the menu lasts only until the remote restarts, and the Wi-Fi screen says so. The password is stored as plain text on the card.

Safety and care

- **Never look into the LED head while it is sending.** 66 infrared LEDs at up to 100 mA each can hurt your eyes at short range, and you cannot see the light. A phone camera shows it as purple.
- **Carrying it is safe.** The AA pack is disconnected from the LEDs unless the remote is sending a code. Keys pressed in a bag still send codes, so take the cells out for long trips.
- **Cells:** use four matching NiMH cells, all charged together. Never mix old and new, or NiMH and alkaline. Remove flat cells.

- **The door screw** is the only one you touch. Don't open the rest of the case unless you are repairing it.
- **Cleaning:** wipe with a dry or slightly damp cloth. Keep water out of the case, the slots and the keys.

Troubleshooting

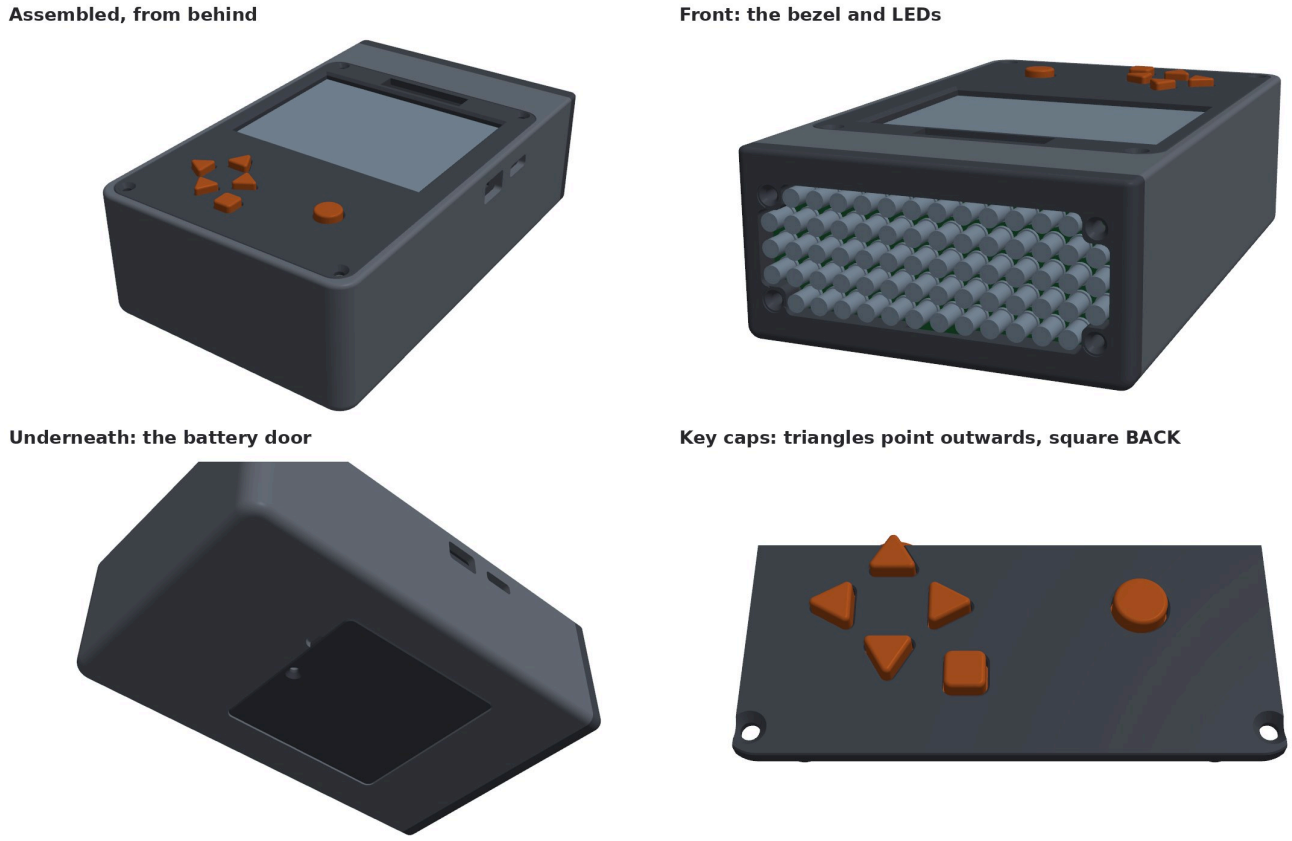
What you see	Likely cause	What to do
Header says `AA out`, or the footer says "LEDs off - no AA cells in"	No cells in, or they are not making contact	Check four cells are in the holder, the right way round
Key presses do nothing to the device	Wrong code set, or the device is out of line	Use Code search with the device's Power key. Aim the head straight at its sensor
Short range, header says `LOW`	AA cells are nearly flat	Swap in charged cells. Below 4.6 V the remote asks for them
Header says `scr` low, or the screen is blank	The screen's 18650 is flat	Charge it through the USB-C port
Screen says the SD card or database is missing	No card, wrong format, or no `irdb.bin`	Use a FAT32 card with `irdb.bin` in its root
Status screen says a key is held since power-up	A stuck key, or a short on the ribbon	Free the key and restart. If it stays, check the ribbon at both ends
Web page won't open	The remote is asleep, or you are on the wrong network	Press a key, join LR-Remote, and open http://192.168.4.1
Wi-Fi screen says "wrong password?" or "network not visible"	A mistyped password or name, or a 5 GHz-only network	Menu → Wi-Fi network: pick the network from the list rather than typing its name, then retype the password (it is case-sensitive). The remote only sees 2.4 GHz networks
Code update refused	The file was built for a different firmware version	Use the `irdb.bin` that came with this firmware. The old one is kept as `irdb.old`

What you see	Likely cause	What to do
No key clicks	Sound is muted, or the speaker's codec didn't answer at start-up	Check Menu → Sound. If it says "no codec found", turn the screen off and on again with its PWR button. The keys and IR work either way

Menu → Status shows both battery voltages, the SD card, the database and any stuck key.

Building the housing

The housing is four printed parts plus six key caps, held together with M3 screws that cut their own threads. It is one 107 × 52 × 170 mm block: the bezel at the front, the body (LED cavity and tray in one piece), the lid on top, and a battery door underneath. The print files are in `housing/stl_lite/`, already turned the way they print.



The finished housing: from behind, from the front, underneath, and the key caps in the lid.

What to print

Part	How it sits on the bed	Size (mm)
Bezel	Front face down, posts up	107.4 × 52 × 8.7

Part	How it sits on the bed	Size (mm)
Body	Floor down	107.4 × 163.8 × 52
Lid	Top face down	100.9 × 138.5 × 17.7
Battery door	Inside face (tongue) down	70 × 75.6 × 3
D-pad key cap	Base down, **print 4**	13.2 Ø × 9.7
SELECT key cap	Base down	14 Ø × 9.7
BACK key cap (square)	Base down	13.2 Ø × 9.7

Every part fits a Prusa MINI (180 × 180 × 180 mm). Use **PETG** if you can: it stays stiff in a hot car and handles the snap hooks well. PLA is fine for indoor use. Print at 0.2 mm layers with 3 perimeters and 15–20 % infill.

Supports: only under the battery bay roof. Inside the body, the bay's roof spans 59 mm, too long to bridge cleanly. In PrusaSlicer, right-click the body → Add support enforcer → Load, pick `body_bay_support_enforcer.stl` (it lines up by itself), and set supports to *For support enforcers only*. The bay opens onto the bed, so the supports pull out through the door opening. Everything else prints without support.

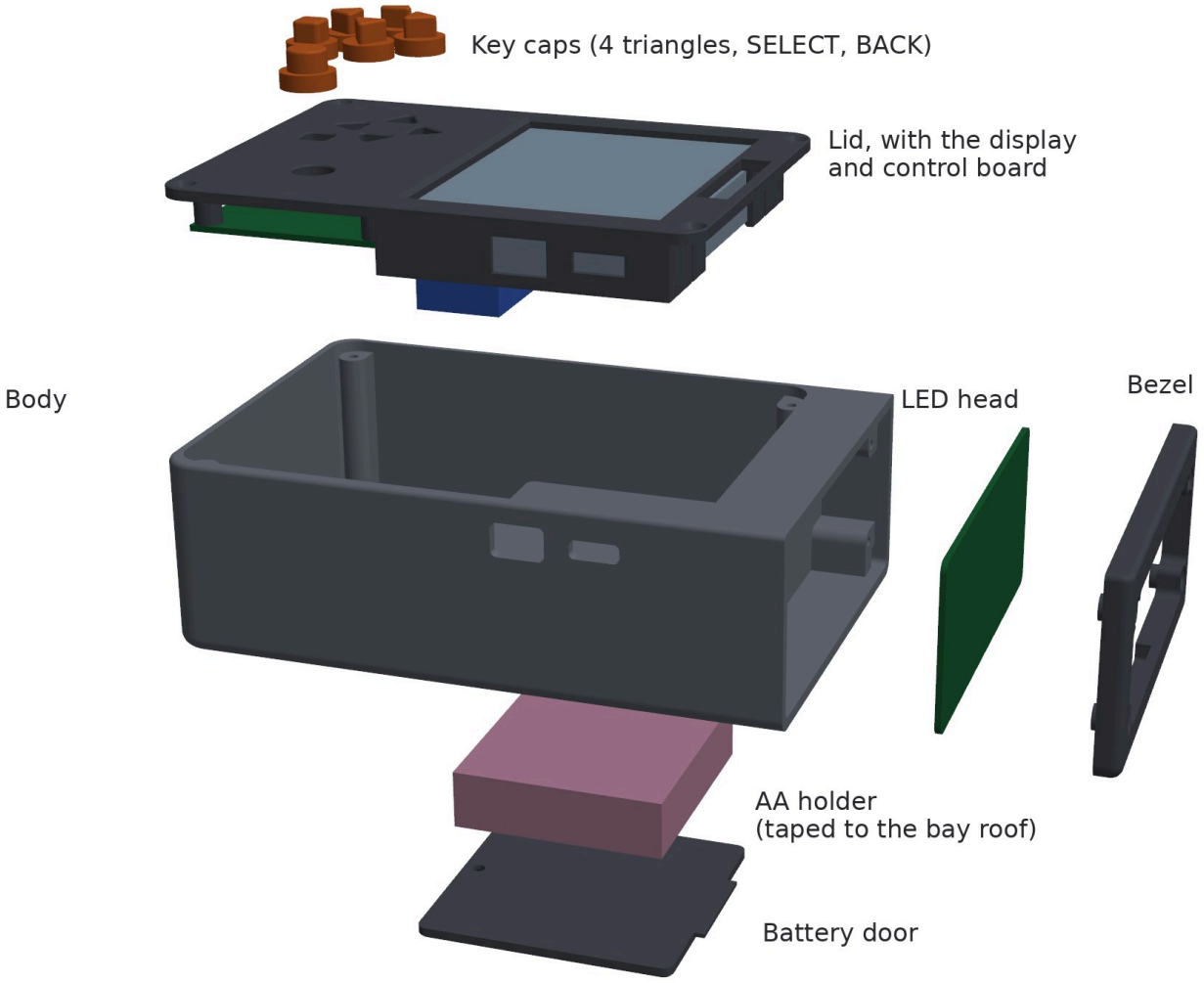
What you need

Qty	Item	Used for
4	M3 × 25 countersunk screw	Bezel → LED head → body
4	M3 × 10 countersunk screw	Lid → body
4	M3 × 8 pan-head screw	Control board → lid
1	M3 × 10 countersunk screw + M3 nut	Battery door
1	Keystone 2478 4 × AA holder	The AA cells
1	JST XH 2-pin plug and 2 crimps	Holder leads → control board J1
—	Double-sided foam tape, about 1 mm	Holds the AA holder to the bay roof
1 + 1	10-way ribbon (6″), and 16-way ribbon (12″, cut to about 70 mm) with one spare 16-way IDC socket for its new end	LED head → control board, control board → display
1	Double-sided 0.1″ pin header, two 8-pin pieces (two 5-pin pieces from the same strip are the LED head's J1)	Turns the 16-way ribbon's socket into a plug for the display's header

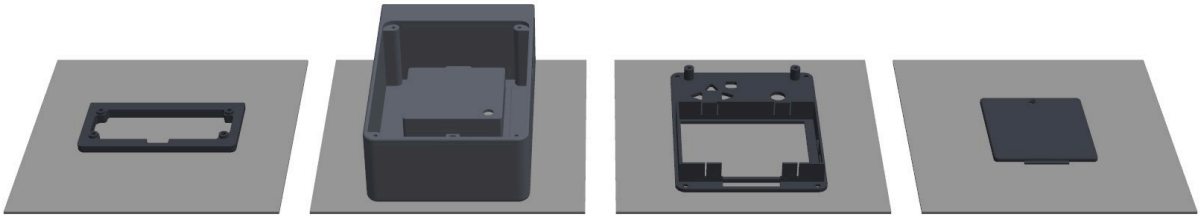
The screws are self-tapping into printed pilot holes, so no inserts are needed. Go easy on the last turn in PLA.

Putting it together

Exploded: how the parts go together



As printed: bezel, body, lid, door (each in its STL orientation)



How the parts go together (top) and how each one sits on the printer bed (bottom).

1. **Battery holder.** Crimp the XH plug onto the holder's leads: red (+) to pin 1, black to pin 2. Feed the leads up through the hole in the bay roof from underneath, then stick the holder to the roof with foam tape, open side down. Drop the M3 nut into the hex pocket behind the bay, from inside the tray.
2. **LED head.** Feed the 10-way ribbon through the opening in the LED cavity's back wall, from the tray side. Plug it into the LED head's J1 (pin 1 marked) with the board held in front of the cavity, J1 at the **bottom**. Slide the board into the cavity, LEDs facing out,

then fit the bezel and the four M3 × 25 screws. Tighten until the bezel seats and the board stops rattling, no further.

3. **Lid.** Drop the key caps into their holes from underneath: the four triangles point outwards (UP towards the LED head), SELECT on the right, and the square BACK at the D-pad's lower right. Screw the control board to the four bosses with the M3 × 8 screws, keys facing the caps. Push the display into its sleeve from underneath until the four hooks click over its back plate: glass to the window, USB-C on the **right**, image top towards the LED head.
4. **Cables.** First shorten the 16-way ribbon: both its ends are in the lid, about 50 mm apart along its route, and there is no room inside for the full 30 cm. Cut it so about 70 mm remains between the socket centres, and press the spare IDC socket onto the cut end in a vice, between two scraps of wood. Copy the socket at the other end: red stripe at the same end, polarising key on the same face of the ribbon. Check with a multimeter that pin 1 goes to pin 1 and pin 16 to pin 16. Then plug it from the control board's J2 to the display header (red stripe on pin 1: looking at the display board from the back with its 18650 at the bottom, pin 1 (3V3) is the bottom-left pin, with pin 2 (VBUS) above it; the socket isn't keyed, so check this before plugging in). Plug the AA holder's plug into J1, and the 10-way ribbon into J5. Where the ribbons pass under the display's 18650 cell there is about 3 mm, so lay them flat and side by side: the 10-way on the left half of the bay roof (towards J5) and the 16-way on the right half (towards J2). Close the lid with the four M3 × 10 screws.
5. **Cells.** Charge four NiMH cells, put them in the holder, and fit the door: tongue in first at the front, then swing the back up and fit the M3 × 10 screw.

Check before the first power-on: with the cells in, the remote's header should show `AA nn%`. If it shows `AA out`, check the holder plug's polarity (red to pin 1) before going further. The board has no reverse-polarity protection.

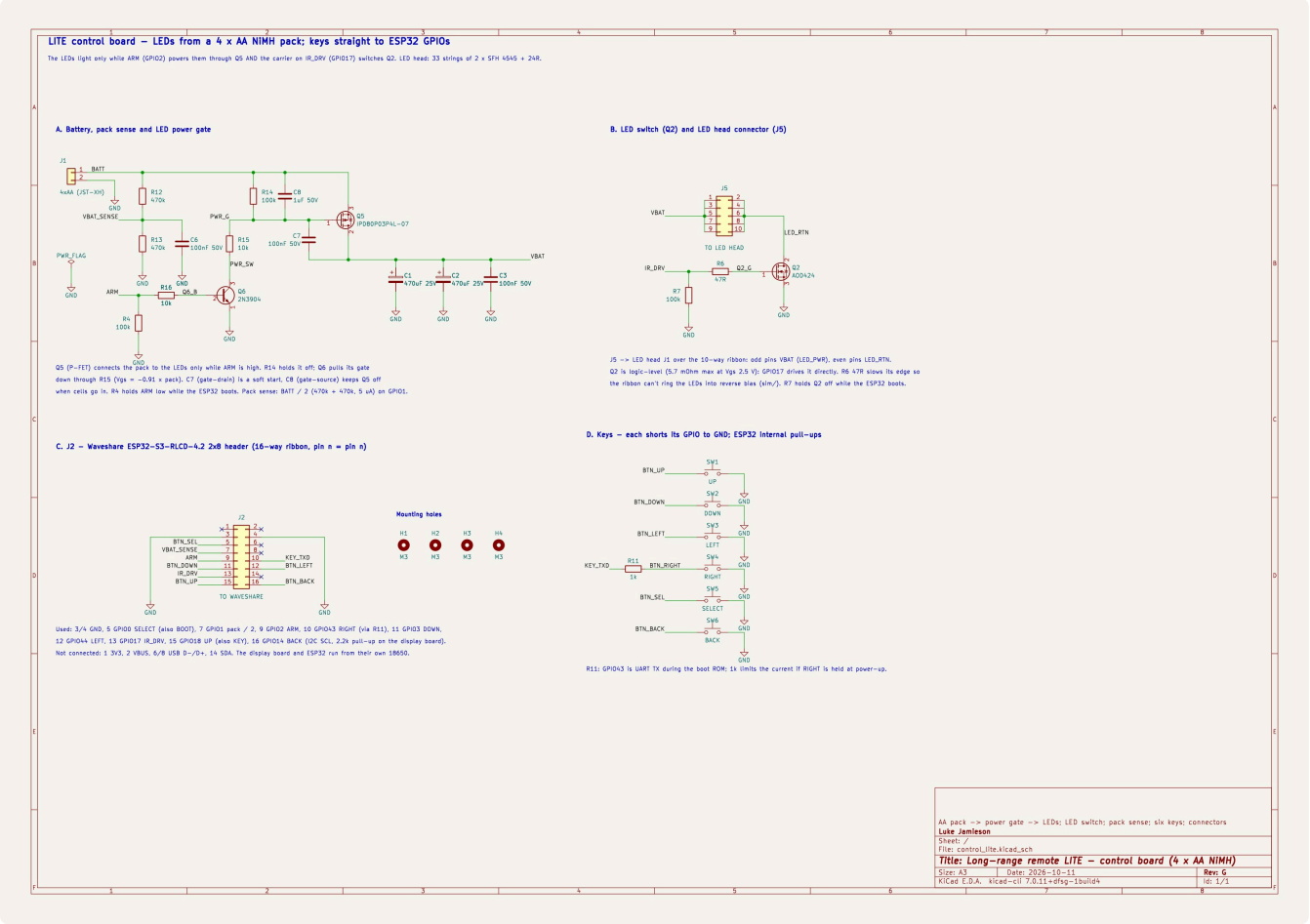
Boards and electronics

The remote has two circuit boards of its own, plus the Waveshare display board. The **control board** (82 × 52 mm) sits under the lid with the six keys facing up through it. The **LED head** (99.8 × 44.8 mm) sits behind the bezel. A 10-way ribbon joins them, and a 16-way ribbon runs from the control board to the display's header. The full KiCad projects, printable PDFs and Gerbers are in the project files.

Control board

The AA cells connect at J1. Q5, a P-channel MOSFET, connects them to the LEDs only while the remote is sending. The ESP32 switches it on through Q6 (ARM, GPIO2), and C7 and C8 make it turn on gently. Q2, a logic-level MOSFET that switches fully on at 2.5 V,

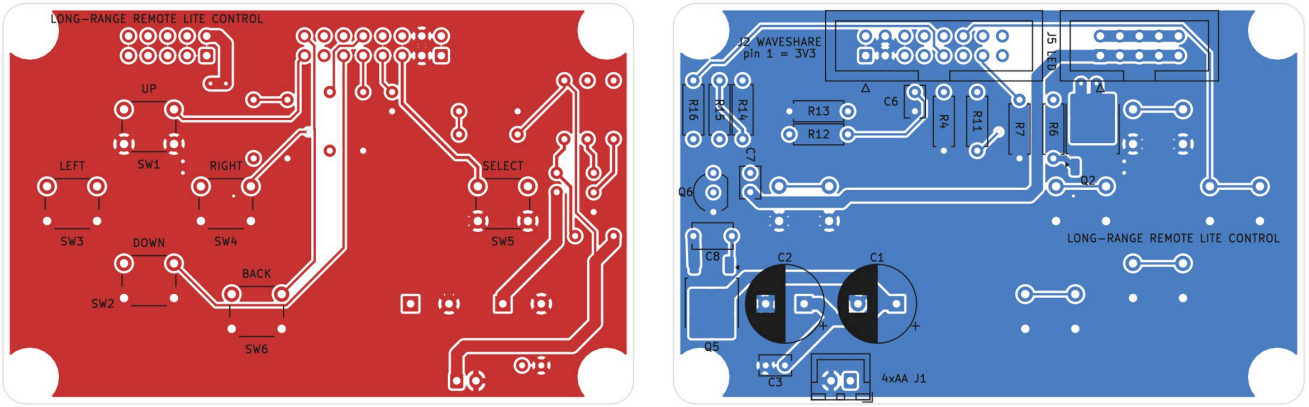
pulses the LEDs at the IR carrier frequency, driven straight from GPIO17 through R6. R6 is 47 Ω on purpose: it slows Q2's switching edge so the ribbon can't ring and reverse-bias the LEDs. R12 and R13 halve the pack voltage for the ESP32 to read on GPIO1. Each key connects its own ESP32 pin to ground. SELECT is on GPIO0, the ESP32's boot pin, so holding SELECT while the remote starts puts it in its bootloader; UP shares GPIO18 with the display board's own KEY button, so that button works as UP. BACK is on GPIO14, the display board's I2C clock line: nothing else uses that bus, and the display board's own pull-up holds the pin high.



Control board schematic, in four blocks: the battery, pack sense and LED power gate (Q5, Q6) top left; the LED switch Q2 and the LED head connector J5 top right; the display connector J2 bottom left; and the six keys bottom right.

Front (keys only)

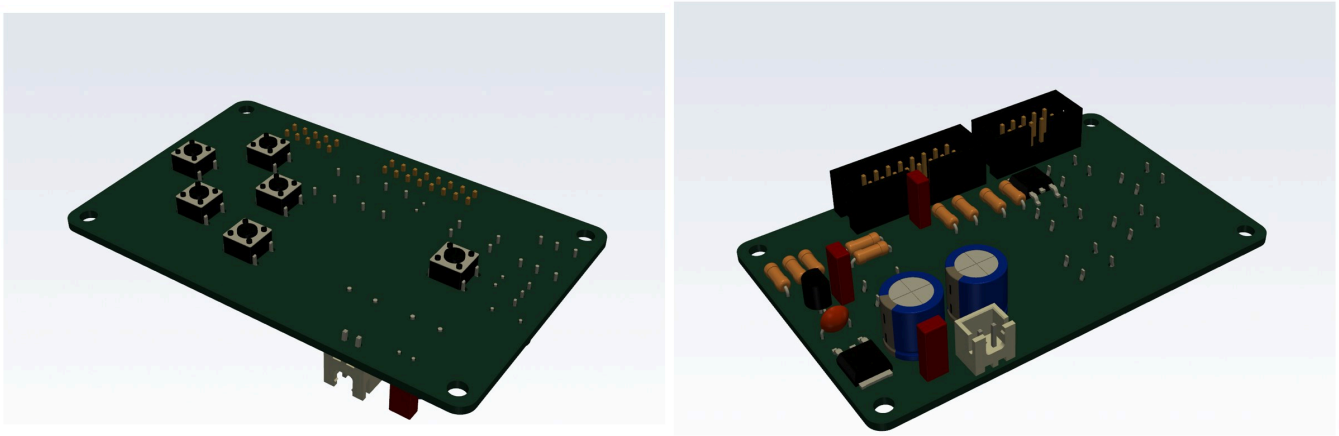
Back



Control board layout. Front (facing up through the lid): the six keys and nothing else, with the D-pad on the left and SELECT on the right, and BACK at the D-pad's lower right. Back: everything else, including the two ribbon connectors (J5 to the LED head, J2 to the display), the LED switch Q2 (a surface-mount DPAK right beside J5), the 470 µF capacitors, the AA battery plug J1, and the power gate (Q5, Q6 and their resistors and capacitors) at the left-hand end.

Front

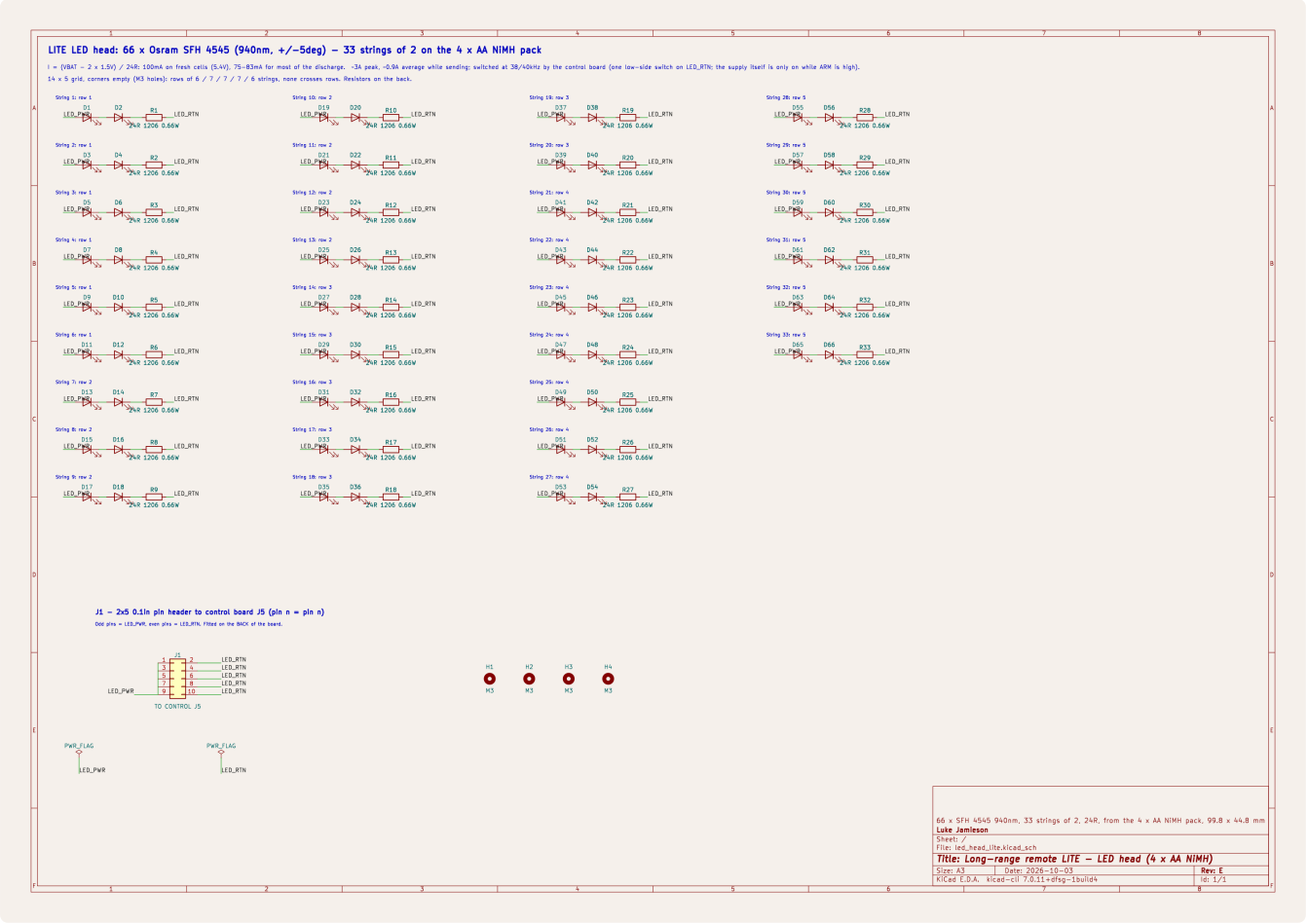
Back



Control board in 3D, front and back.

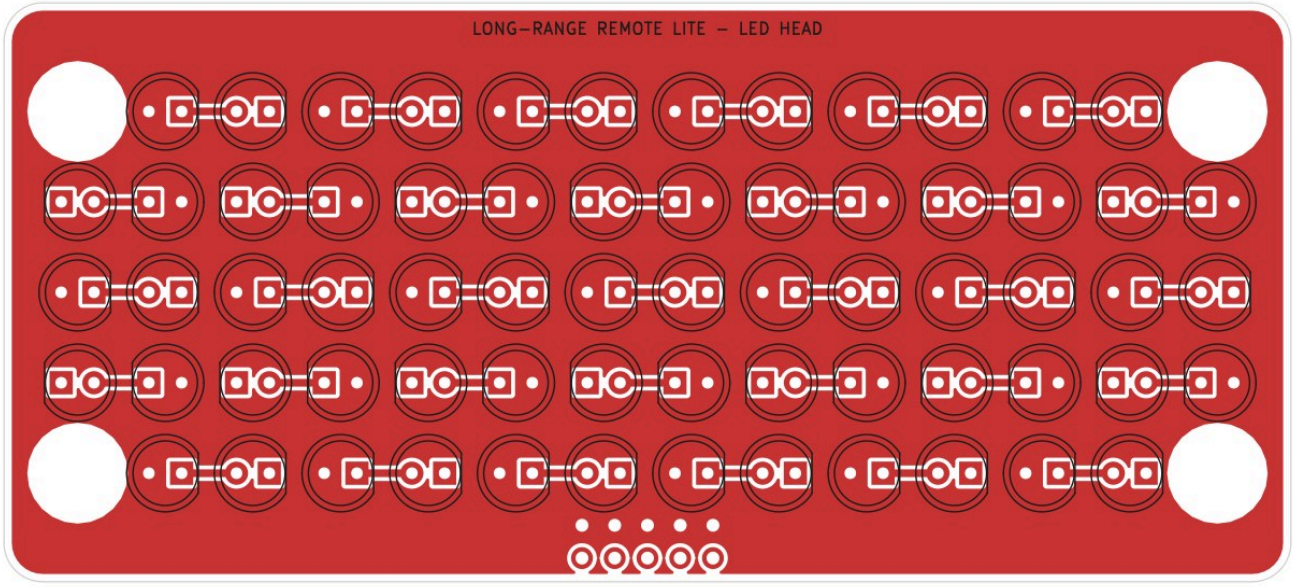
LED head

66 SFH 4545 infrared LEDs in 33 strings of two, each with its own 24 Ω resistor. The LEDs are on the front and the resistors and J1 on the back. The front copper is the supply (LED_PWR) for every string, and the back copper is the shared return (LED_RTN) to the control board.

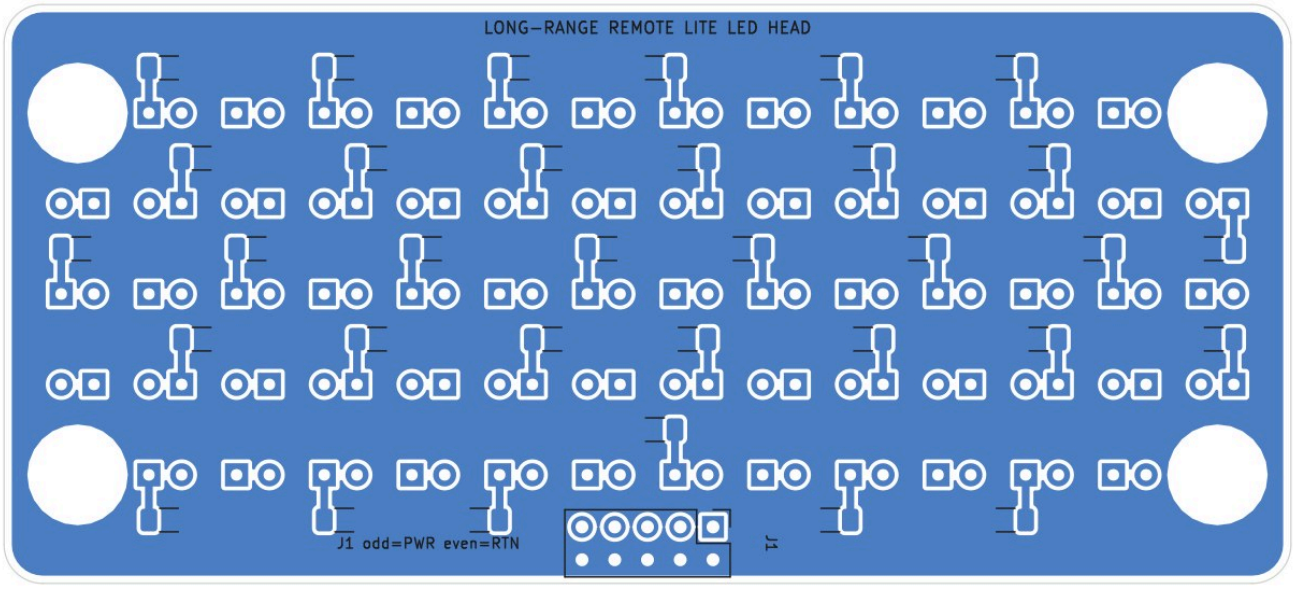


LED head schematic: 33 identical strings, and J1 (odd pins supply, even pins return).

Front (LEDs)



Back



LED head layout. Front: the 14 × 5 LED grid, with the corners left free for the four M3 screws. Back: the 33 string resistors between the LED rows, and J1 along the bottom edge.

Image — LED head 3D views, front and back

LED head in 3D, front and back.

The 3D views come from KiCad's 3D models: the LEDs show as KiCad's generic red 5 mm LED, and the copper and silkscreen are not drawn. The board layouts above show those.